ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Exploring Forces** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 4 N	Balanced forces.
2	**(c)** positive charge	Like charges repel.
3	**(d)** change in colour	Force does not change colour.
4	**(b)** 8 N	20 - 12 = 8 N.
5	**(d)** it induces opposite charge near it	Charge separation.
6	**(b)** 0.3 kg	3/10.
7	**(a)** 0 N	Balanced.
8	**(d)** charge flows to earth	Grounded.
9	**(d)** have two poles	Poles always occur in pairs.
10	**(a)** 5 N upward	Balanced forces.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Net = 15 - 6 = 9 N (1); In the direction of push (1).	1+1
12	Spikes increase friction with the ground (1); Prevents slipping (1).	1+1
13	Friction between hands (1); Energy changes to heat (1).	1+1
14	Static friction exceeds sliding friction (1); Starting needs larger force (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	25 N is static friction at the limit (1); 18 N is sliding friction (1); Static friction is larger than sliding friction (1).	1+1+1
16	Moon's gravity is less (1); Weight is gravitational force (1); Mass is the amount of matter (1).	1+1+1
17	At the poles (1); Strongest magnetic force is at the poles (1); Few at the middle (1).	1+1+1
	OR: Same direction: 20 + 12 = 32 N (1); Opposite directions: 20 - 12 = 8 N (1); Net force depends on direction (1).	1+1+1
18	Pull a block with a spring balance (1); Add weights on the block (1); Reading increases (1).	1+1+1
	OR: Larger air resistance on leaf (1); Leaf has a large area, little weight (1); Air resistance matters (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Readings increase with roughness (1); Friction depends on nature of surface (1); Carpet has the highest (1); Adding mass increases normal force (1); Readings increase (1).	1+1+1+1+1
	OR: Feather: large air resistance (1); Stone: small compared with weight (1); Parachutist: opens a parachute (1); Air resistance grows with speed (1); Reaches constant speed (1).	1+1+1+1+1
20	(a) Static friction (1); (b) Sliding friction (1); (c) Rolling friction (1); (d) Static friction (1); Rolling is the smallest (1).	1+1+1+1+1
	OR: Attracts iron (1); Like poles repel, unlike attract (1); Compass is a small magnet (1); Points north-south (1); Earth is a large magnet (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that friction acts in the direction of motion; it acts **opposite** to the motion (or its tendency).
- Using kg as the unit of weight, or saying weight is constant everywhere.
- Subtracting forces that act in the same direction, or adding forces that act in opposite directions.
- Thinking friction is always harmful; it is needed for walking, brakes and gripping.
- Saying magnetic and electrostatic forces need contact, or that electrostatic force only repels.
- Concluding that an object not moving has no force on it; balanced forces also give zero motion change.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Exploring Forces** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** open	Induction.
2	**(d)** a non-contact force	Acts without touching.
3	**(a)** friction is smaller on the smooth floor	Smooth surface has less friction.
4	**(d)** Unlike charges attract	Law of charges.
5	**(d)** like poles face each other	Repulsion.
6	**(b)** 10 N	Add them.
7	**(d)** water reduces friction	Less grip.
8	**(d)** friction	Opposes motion.
9	**(b)** 2 N to the right	Subtract.
10	**(a)** increase friction	Better grip.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	To reduce air and water resistance (1); Helps move faster with less force (1).	1+1
12	They change sliding friction into rolling friction (1); Wheel turns easily (1).	1+1
13	Charge passes to both leaves (1); They carry the same charge and repel (1).	1+1
14	They push each other away (1); Like poles repel (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Glass rod becomes positive (1); Plastic rod becomes negative (1); Unlike charges attract (1).	1+1+1
16	Ski should slide (1); Tyre needs grip (1); Friction is reduced and increased according to need (1).	1+1+1
17	Reduce: machines, skis, bicycle chain (1); Increase: brakes, tyres, shoes (1); Use oil or grip (1).	1+1+1
	OR: Charge from cloud reaches rod (1); Pointed rod helps the charge flow easily (1); Earth wire takes charge to the ground (1).	1+1+1
18	Metal parts and leaves hold charge (1); Hand provides a path to earth (1); Charge flows out (1).	1+1+1
	OR: Rubbing synthetic cloth charges it (1); Clothes cling together or crackle (1); Use softener or humid air (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Rubbing transfers charge (1); Comb gets charge (1); Paper gets induced charge (1); Attracted to comb (1); Paper gets same charge after touching and is repelled (1).	1+1+1+1+1
	OR: Walking, brakes, matches, knots, writing (2); Oil, powder, bearings, polish, streamlining (2); Both needed for efficiency (1).	2+2+1
20	Gravity: falling objects, orbits (1); Magnetic: compass, door catches (1); Electrostatic: lightning, photocopier (1); All non-contact (1); Acts at a distance (1).	1+1+1+1+1
	OR: Smooth box moves faster (1); Rough box moves slower or not at all (1); More friction on rough box (1); Net force is less (1); Friction opposes the push (1).	1+1+1+1+1

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- Subtracting forces that act in the same direction, or adding forces that act in opposite directions.
- Thinking friction is always harmful; it is needed for walking, brakes and gripping.
- Saying magnetic and electrostatic forces need contact, or that electrostatic force only repels.
- Concluding that an object not moving has no force on it; balanced forces also give zero motion change.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Exploring Forces** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** balanced	Weight equals drag.
2	**(c)** friction and air resistance	Opposing forces.
3	**(b)** fluid friction (drag)	Smooth flow.
4	**(a)** a heavy box on a rough floor	Large load and rough.
5	**(b)** less	Moon's gravity is smaller.
6	**(a)** increase friction with the rim	Grip.
7	**(d)** petrol or gas leaks	May spark.
8	**(b)** 2 kg	Mass is constant.
9	**(a)** rolling friction is less than sliding friction	Rolling is easier.
10	**(c)** crouch low and keep feet together	Reduces risk.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Moisture helps charge leak to the ground (1); Water is a conductor (1).	1+1
12	Rubbing charges the comb (1); Charged comb attracts neutral paper (1).	1+1
13	60 / 6 (1); = 10 N (1).	1+1
14	Friction needs contact; gravity does not (1); Friction opposes motion; gravity pulls towards Earth (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Nature of surfaces (1); Force pressing them together (1); Not much on the area of contact (1).	1+1+1
16	Moves: hits carrom coin (1); Stops: catching a ball (1); Changes direction: bat hits ball (1).	1+1+1
17	Needle behaves like a magnet (1); Earth acts as a magnet (1); It aligns north-south (1).	1+1+1
	OR: Large area increases air resistance (1); Air resistance balances weight partly (1); Speed becomes small (1).	1+1+1
18	Equal pull: no motion (1); Unequal: unbalanced force (1); Net force towards stronger team (1).	1+1+1
	OR: Magnetic: acts on magnetic materials; electrostatic on charged bodies (1); Magnet has poles; charges positive and negative (1); Both act at a distance (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Balanced forces: constant velocity or rest (1); Net force zero (1); 25 N push: net 10 N (1); Box speeds up (1); In the direction of push (1).	1+1+1+1+1
	OR: Charges flow to nearest point (1); Tall objects are closer to the cloud (1); Stay indoors (1); Avoid open fields and water (1); Switch off appliances (1).	1+1+1+1+1
20	Metal disc, rod and leaves (1); Charged body touches disc (1); Leaves diverge (1); Touch with a finger to earth it (1); Leaves collapse (1).	1+1+1+1+1
	OR: Drop a stone and a paper crumpled (1); Both land almost together (1); Air resistance is small for crumpled paper (1); Gravity accelerates all alike (1); Same fall time (1).	1+1+1+1+1

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